

KERALA DIPLOMA CURRICULUM REVISION 2026

Course 5053A

5 Credits: 4

Program Core

Source-grounded

Automotive Power Train

□ To explain the principle of engine control system and transmission control systems

OFFICIAL SOURCE DECLARATION

How this lesson was prepared

The supplied official SITTTTR syllabus PDF for Course 5053A is the authoritative source for course information, objectives, course outcomes, CO-PO mapping, modules, topics, activities, assessment structure and references. Explanations, study prompts and Malayalam support are educational elaboration and are not presented as additional official syllabus text.

Source file: 5053A.pdf from the uploaded official SITTTTR syllabus archive.

Course code and title: preserved from the source.

COURSE DASHBOARD

Official course information

Course information

Item	Official information
Programme	Diploma in Automobile Engineering
Course code	5053A
Course title	Automotive Power Train
Semester	5 Credits: 4
Course category	Program Core
Periods per week	4 (L:4, T:0, P:0) Periods per semester: 60
Periods per semester	60
Credits	4
Prerequisite	See the official syllabus PDF
Assessment	Use the official assessment section reproduced below

Modules

4 official module sections detected.

Topic entries

13 source-aligned topic entries retained.

Study mode

Theory and application emphasis.

STUDY METHOD**How to use this lesson****First pass**

Read the official source declaration, dashboard and module transcript. Mark unfamiliar terms without changing the official terminology.

Second pass

Use each topic card to build a definition, principle, steps or components, application and exam answer. Attempt the Q&A before opening the answers.

Practical evidence

Where the course is laboratory or workshop based, maintain an observation notebook with procedure, instruments, readings, safety points and conclusion.

Revision pass

Return to the source excerpt, coverage table, activities and model paper. The generated paper is practice material, not an official examination paper.

LEARNING OUTCOMES

Objectives, outcomes and mapping

Course objectives

- To explain the principle of engine control system and transmission control systems

Course outcomes

CO	Official outcome	Study level
CO1	14 Understanding Systems Identify various sensors and actuators used in	See official syllabus
CO2	15 Applying electronic engine control system.	See official syllabus
CO3	Outline the features of digital engine control system 15 Understanding Outline the integrated engine control system and	See official syllabus
CO4	14 Understanding automatic transmission control	See official syllabus

CO-PO mapping evidence

Row	Extracted official mapping
CO1	CO1 3 2
CO2	CO2 3 2
CO3	CO3 3 2
CO4	CO4 2 2

COVERAGE AUDIT

Complete syllabus coverage

Every detected official module is represented below. Each module contains topic cards and an expandable official syllabus transcript to preserve source terminology.

Module coverage

Module	Official heading	Topic entries	Hours
Module 1	Summarize the concept of electronic engine control Systems	3	As specified
Module 2	system.	4	As specified
Module 3	Outline the features of digital engine control system	3	As specified
Module 4	control	3	As specified

MODULE 1

Summarize the concept of electronic engine control Systems

This module is presented from the official syllabus scope for Automotive Power Train. Use the topic cards for learning and the source transcript for exact coverage.

As specified
official hours

CO-linked
see outcomes

Learning goals

- Explain the official Module 1 scope and its relationship to the course outcomes.
- Recognise the terminology, sequence, components and evidence expected in examination answers.
- Connect at least one official topic to a practical, industrial, laboratory or design context.

Key facts

- Official module title: Summarize the concept of electronic engine control Systems
- Official duration: As specified
- Official topic entries captured: 3
- The full source excerpt is preserved below for coverage checking.

▼ Outline the functions of engine and controls

Outline the functions of engine and controls is an official topic in Module 1 of Automotive Power Train. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Outline the functions of engine and controls using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, outline the functions of engine and controls should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Outline the functions of engine and controls means in the official course context

Aspect	What to prepare
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 1-ഓൺ ഔട്ട്ലൈൻ എൻജിൻ നിയന്ത്രണങ്ങൾ നിർവ്വചന/അർത്ഥം നിർവ്വചിക്കുക. ഏതെങ്കിലും ഏതെങ്കിലും, steps, application നിർവ്വചിക്കുക നിർവ്വചിക്കുക. Technical terms English-ൽ നിർവ്വചിക്കുക നിർവ്വചിക്കുക.

▼ Outline the terms related to engine performance

Outline the terms related to engine performance is an official topic in Module 1 of Automotive Power Train. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Outline the terms related to engine performance using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, outline the terms related to engine performance should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Outline the terms related to engine performance means in the official course context

Aspect	What to prepare
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 1-ഓൺ ഔട്ട്ലൈൻ തെർമ്സ് റിലേറ്റഡ് ടു എൻജിൻ പെർഫോമൻസ് ഓൺ ഓൺ ഓൺ definition/meaning ഓൺ ഓൺ ഓൺ. ഓൺ ഓൺ ഓൺ, steps, application ഓൺ ഓൺ ഓൺ. Technical terms English-ഓൺ ഓൺ ഓൺ.

▼ Explain the mapping of engine. 3 Understanding M1. 04 Outline electronic fuel control system 4 Understanding The concept of electronic Engine control system - Engine functions and control - Parameters and variables - Input to controller - output from controller - Definition of engine performance terms - Power - torque - fuel consumption - Volumetric efficiency - Thermal efficiency - calibration, Engine mapping-Effect of air fuel ratio, spark timing and exhaust gas recirculation on performance - Electronic fuel control system - Engine control sequence - open loop control - closed loop control - closed loop operation - Idle speed control. Identify various sensors and actuators used in electronic engine control

Explain the mapping of engine. 3 Understanding M1. 04 Outline electronic fuel control system 4 Understanding The concept of electronic Engine control system - Engine functions and control - Parameters and variables - Input to controller - output from controller - Definition of engine performance terms - Power - torque - fuel consumption - Volumetric efficiency - Thermal efficiency - calibration, Engine mapping-Effect of air fuel ratio, spark timing and exhaust gas recirculation on performance - Electronic fuel control system - Engine control sequence - open loop control - closed loop control - closed loop operation - Idle speed control. Identify various sensors and actuators used in electronic engine control is an official topic in Module 1 of Automotive Power Train. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Explain the mapping of engine. 3 Understanding M1. 04 Outline electronic fuel control system 4 Understanding The concept of electronic Engine control system - Engine functions and control - Parameters and variables - Input to controller - output from controller - Definition of engine performance terms - Power - torque - fuel consumption - Volumetric efficiency - Thermal efficiency - calibration, Engine mapping-Effect of air fuel ratio, spark timing and exhaust gas recirculation on performance - Electronic fuel control system - Engine control sequence - open loop control - closed loop control - closed loop operation - Idle speed control. Identify various sensors and actuators used in electronic engine control using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, explain the mapping of engine. 3 understanding m1. 04 outline electronic fuel control system 4 understanding the concept of electronic engine control system - engine functions and control - parameters and variables - input to controller - output from controller - definition of engine performance terms - power - torque - fuel consumption - volumetric efficiency - thermal efficiency - calibration, engine mapping-effect of air fuel ratio, spark timing and exhaust gas recirculation on performance - electronic fuel control system - engine control sequence - open loop control - closed loop control - closed loop operation - idle speed control. identify various sensors and actuators used in electronic engine control should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Explain the mapping of engine. 3 Understanding M1. 04 Outline electronic fuel control system 4 Understanding The concept of electronic

Aspect	What to prepare
	Engine control system - Engine functions and control - Parameters and variables - Input to controller - output from controller - Definition of engine performance terms - Power - torque - fuel consumption - Volumetric efficiency - Thermal efficiency - calibration, Engine mapping- Effect of air fuel ratio, spark timing and exhaust gas recirculation on performance - Electronic fuel control system - Engine control sequence - open loop control - closed loop control - closed loop operation - Idle speed control. Identify various sensors and actuators used in electronic engine control means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 1- Explain the mapping of engine. 3 Understanding M1. 04 Outline electronic fuel control system 4 Understanding The concept of electronic Engine control system - Engine functions and control - Parameters and variables - Input to controller - output from controller - Definition of engine performance terms - Power - torque - fuel consumption - Volumetric efficiency - Thermal efficiency - calibration, Engine mapping-Effect of air fuel ratio, spark timing and exhaust gas recirculation on performance - Electronic fuel control system - Engine control sequence - open loop control - closed loop control - closed loop operation - Idle speed control. Identify various sensors and actuators used in electronic engine control

 definition/meaning
 .

പ്രക്രിയ, ഘട്ടങ്ങൾ, steps, application എന്നിവയെക്കുറിച്ച് വിശദീകരിക്കുക.
Technical terms English-ൽ പദങ്ങൾ വിശദീകരിക്കുക.

► **Official syllabus detail and terminology (expand in digital version)**

Module study routine: Read the source wording, make a one-page concept map, practise one short answer and one structured answer, then review the Malayalam support notes.

MODULE 2**system.**

This module is presented from the official syllabus scope for Automotive Power Train. Use the topic cards for learning and the source transcript for exact coverage.

As specified
official hours

CO-linked
see outcomes

Learning goals

- Explain the official Module 2 scope and its relationship to the course outcomes.
- Recognise the terminology, sequence, components and evidence expected in examination answers.
- Connect at least one official topic to a practical, industrial, laboratory or design context.

Key facts

- Official module title: system.
- Official duration: As specified
- Official topic entries captured: 4
- The full source excerpt is preserved below for coverage checking.

▼ Outline the engine Variables

Outline the engine Variables is an official topic in Module 2 of Automotive Power Train. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Outline the engine Variables using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, outline the engine variables should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Outline the engine Variables means in the official course context
Study lens	Principle → components or stages → result → application

Aspect	What to prepare
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 2-ഓൺ ഔട്ട്ലൈൻ എൻജിൻ വേരിയബിൾസ്

ഓട്ടോമോട്ടീവ് പവർ ട്രെയിൻ definition/meaning ഓട്ടോമോട്ടീവ് പവർ ട്രെയിൻ. ഓട്ടോമോട്ടീവ് പവർ ട്രെയിൻ, steps, application ഓട്ടോമോട്ടീവ് പവർ ട്രെയിൻ. Technical terms English-ൽ ഓട്ടോമോട്ടീവ് പവർ ട്രെയിൻ.

▼ Illustrate the working of various sensors

Illustrate the working of various sensors is an official topic in Module 2 of Automotive Power Train. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Illustrate the working of various sensors using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, illustrate the working of various sensors should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Illustrate the working of various sensors means in the official course context

Aspect	What to prepare
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 2-ഇംഗ്ലീഷ് Illustrate the working of various sensors
 ഇംഗ്ലീഷ് definition/meaning ഇംഗ്ലീഷ്. ഇംഗ്ലീഷ് ഇംഗ്ലീഷ്
 ഇംഗ്ലീഷ്, steps, application ഇംഗ്ലീഷ് ഇംഗ്ലീഷ് ഇംഗ്ലീഷ്. Technical terms
 English-ഇംഗ്ലീഷ് ഇംഗ്ലീഷ് ഇംഗ്ലീഷ്.

▼ Illustrate the working of various actuators

Illustrate the working of various actuators is an official topic in Module 2 of Automotive Power Train. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Illustrate the working of various actuators using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, illustrate the working of various actuators should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Illustrate the working of various actuators means in the official course context

Aspect	What to prepare
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 2-ഇംഗ്ലീഷ് Illustrate the working of various actuators ഇംഗ്ലീഷ് definition/meaning ഇംഗ്ലീഷ്. ഇംഗ്ലീഷ് ഇംഗ്ലീഷ്, steps, application ഇംഗ്ലീഷ് ഇംഗ്ലീഷ് ഇംഗ്ലീഷ്. Technical terms English-ഇംഗ്ലീഷ് ഇംഗ്ലീഷ് ഇംഗ്ലീഷ്.

▼ Identify various actuators and actuators 1 Applying Typical electronic Engine Control System. Variables to be measured, Sensors and actuators- Temperature sensor- Pressure sensor-Mass air flow sensor- Oxygen sensor- Position sensor-Position/speed sensor-Knock sensor- Automotive Engine Control Actuators-Solenoid-switches-relays-motors-Fuel injection- Exhaust gas recirculation actuators-Variable valve timing.

Identify various actuators and actuators 1 Applying Typical electronic Engine Control System. Variables to be measured, Sensors and actuators- Temperature sensor- Pressure sensor-Mass air flow sensor-Oxygen sensor- Position sensor-Position/speed sensor-Knock sensor- Automotive Engine Control Actuators-Solenoid-switches-relays-motors-Fuel injection- Exhaust gas recirculation actuators-Variable valve timing. is an official topic in Module 2 of Automotive Power Train. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Identify various actuators and actuators 1 Applying Typical electronic Engine Control System. Variables to be measured, Sensors and actuators- Temperature sensor- Pressure sensor-Mass air flow sensor- Oxygen sensor- Position sensor-Position/speed sensor-Knock sensor- Automotive Engine Control Actuators-Solenoid-switches-relays-motors-Fuel injection- Exhaust gas recirculation actuators-Variable valve timing. using standard technical terminology.

- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, identify various actuators and actuators 1 applying typical electronic engine control system. variables to be measured, sensors and actuators- temperature sensor- pressure sensor-mass air flow sensor-oxygen sensor- position sensor-position/speed sensor-knock sensor- automotive engine control actuators-solenoid-switches-relays-motors-fuel injection- exhaust gas recirculation actuators-variable valve timing. should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Identify various actuators and actuators 1 Applying Typical electronic Engine Control System. Variables to be measured, Sensors and actuators- Temperature sensor- Pressure sensor-Mass air flow sensor-Oxygen sensor- Position sensor-Position/speed sensor-Knock sensor- Automotive Engine Control Actuators-Solenoid-switches-relays-motors-Fuel injection- Exhaust gas recirculation actuators-Variable valve timing. means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 2-Identify various actuators and actuators
 1 Applying Typical electronic Engine Control System. Variables to be measured, Sensors and actuators- Temperature sensor- Pressure sensor- Mass air flow sensor-Oxygen sensor- Position sensor-Position/speed sensor- Knock sensor- Automotive Engine Control Actuators-Solenoid-switches-relays-motors-Fuel injection- Exhaust gas recirculation actuators-Variable valve timing. പദപരിചയം definition/meaning പദപരിചയം. പദപരിചയം പദപരിചയം, steps, application പദപരിചയം പദപരിചയം. Technical terms English-പദപരിചയം പദപരിചയം.

► **Official syllabus detail and terminology (expand in digital version)**

Module study routine: Read the source wording, make a one-page concept map, practise one short answer and one structured answer, then review the Malayalam support notes.

MODULE 3

Outline the features of digital engine control system

This module is presented from the official syllabus scope for Automotive Power Train. Use the topic cards for learning and the source transcript for exact coverage.

As specified
official hours

CO-linked
see outcomes

Learning goals

- Explain the official Module 3 scope and its relationship to the course outcomes.
- Recognise the terminology, sequence, components and evidence expected in examination answers.
- Connect at least one official topic to a practical, industrial, laboratory or design context.

Key facts

- Official module title: Outline the features of digital engine control system
- Official duration: As specified
- Official topic entries captured: 3
- The full source excerpt is preserved below for coverage checking.

▼ 5 Understanding control systems.

5 Understanding control systems. is an official topic in Module 3 of Automotive Power Train. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 5 Understanding control systems. using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 5 understanding control systems. should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What 5 Understanding control systems. means in the official course context

Aspect	What to prepare
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 3-5 Understanding control systems.
 പരീക്ഷയ്ക്കായി definition/meaning നൽകുക. പരീക്ഷയിൽ
 പരീക്ഷിക്കുക, steps, application നൽകുക. Technical terms
 English-ൽ നൽകുക.

▼ Explain various modes for fuel control

Explain various modes for fuel control is an official topic in Module 3 of Automotive Power Train. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Explain various modes for fuel control using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, explain various modes for fuel control should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Explain various modes for fuel control means in the official course context

Aspect	What to prepare
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 3- Explain various modes for fuel control
 പരമ്പരാഗതമായി പദപരിചയം നൽകുന്നു. പദപരിചയം പദപരിചയം
 പദപരിചയം, steps, application പദപരിചയം പദപരിചയം പദപരിചയം. Technical terms
 English- പദപരിചയം പദപരിചയം പദപരിചയം.

▼ Explain Electronic ignition control 5 Understanding Digital power train control systems - features - Components - Digital engine control system diagram - Modes for fuel control - Control during Engine crank - Engine Warm up - After the engine warm up - open loop - closed loop - Acceleration enrichment - Deceleration leaning - Idle speed control - EGR Control - Variable valve timing control - Electronic ignition control- Distributor less Ignition control. Outline the integrated engine control system and automatic transmission

Explain Electronic ignition control 5 Understanding Digital power train control systems - features - Components - Digital engine control system diagram - Modes for fuel control - Control during Engine crank - Engine Warm up - After the engine warm up - open loop - closed loop - Acceleration enrichment - Deceleration leaning - Idle speed control - EGR Control - Variable valve timing control - Electronic ignition control- Distributor less Ignition control. Outline the integrated engine control system and automatic transmission is an official topic in Module 3 of Automotive Power Train. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Explain Electronic ignition control 5 Understanding Digital power train control systems - features - Components - Digital engine control system diagram - Modes for fuel control - Control during Engine crank - Engine Warm up - After the engine warm up - open loop - closed loop - Acceleration enrichment - Deceleration leaning - Idle speed control - EGR

Control - Variable valve timing control - Electronic ignition control-Distributor less Ignition control. Outline the integrated engine control system and automatic transmission using standard technical terminology.

- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, explain electronic ignition control 5 understanding digital power train control systems - features - components - digital engine control system diagram - modes for fuel control - control during engine crank - engine warm up - after the engine warm up - open loop - closed loop - acceleration enrichment - deceleration leaning - idle speed control - egr control - variable valve timing control - electronic ignition control-distributor less ignition control. outline the integrated engine control system and automatic transmission should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Explain Electronic ignition control 5 Understanding Digital power train control systems - features - Components - Digital engine control system diagram - Modes for fuel control - Control during Engine crank - Engine Warm up - After the engine warm up - open loop - closed loop - Acceleration enrichment - Deceleration leaning - Idle speed control - EGR Control - Variable valve timing control - Electronic ignition control-Distributor less Ignition control. Outline the integrated engine control system and automatic transmission means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 3- Explain Electronic ignition control 5 Understanding Digital power train control systems - features - Components - Digital engine control system diagram - Modes for fuel control - Control during Engine crank - Engine Warm up - After the engine warm up - open loop - closed loop - Acceleration enrichment - Deceleration leaning - Idle speed control - EGR Control - Variable valve timing control - Electronic ignition control-Distributor less Ignition control. Outline the integrated engine control system and automatic transmission definition/meaning definition/meaning. steps, application. Technical terms English- Malayalam.

► **Official syllabus detail and terminology (expand in digital version)**

Module study routine: Read the source wording, make a one-page concept map, practise one short answer and one structured answer, then review the Malayalam support notes.

MODULE 4**control**

This module is presented from the official syllabus scope for Automotive Power Train. Use the topic cards for learning and the source transcript for exact coverage.

As specified
official hours

CO-linked
see outcomes

Learning goals

- Explain the official Module 4 scope and its relationship to the course outcomes.
- Recognise the terminology, sequence, components and evidence expected in examination answers.
- Connect at least one official topic to a practical, industrial, laboratory or design context.

Key facts

- Official module title: control
- Official duration: As specified
- Official topic entries captured: 3
- The full source excerpt is preserved below for coverage checking.

▼ Explain integrated engine control systems 4 Understanding Explain improvements in Electronic engine

Explain integrated engine control systems 4 Understanding Explain improvements in Electronic engine is an official topic in Module 4 of Automotive Power Train. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Explain integrated engine control systems 4 Understanding Explain improvements in Electronic engine using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, explain integrated engine control systems 4 understanding explain improvements in electronic engine should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Explain integrated engine control systems 4 Understanding Explain improvements in Electronic engine means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 4- Explain integrated engine control systems 4 Understanding Explain improvements in Electronic engine definition/meaning . . . , steps, application . Technical terms English- .

▼ 5 Understanding control system Explain automatic transmission and traction

5 Understanding control system Explain automatic transmission and traction is an official topic in Module 4 of Automotive Power Train. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 5 Understanding control system Explain automatic transmission and traction using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 5 understanding control system explain automatic transmission and traction should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
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Aspect	What to prepare
Definition/identity	What 5 Understanding control system Explain automatic transmission and traction means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 4-5 Understanding control system Explain automatic transmission and traction ഓട്ടോമറ്റിക് ട്രാൻസ്മിഷൻ, ട്രാക്ഷൻ definition/meaning ഓട്ടോമറ്റിക് ട്രാൻസ്മിഷൻ, ട്രാക്ഷൻ, steps, application ഓട്ടോമറ്റിക് ട്രാൻസ്മിഷൻ, ട്രാക്ഷൻ. Technical terms English-ഓട്ടോമറ്റിക് ട്രാൻസ്മിഷൻ, ട്രാക്ഷൻ.

▼ **5 Understanding control Integrated engine control system - Secondary air Management - Evaporative emission canister purge - Automatic system adjustment - System diagnosis - Improvements in Electronic Engine Control system - Oxygen sensor improvements - Heated exhaust gas oxygen sensor - Fuel injection Timing - Transmission control - Automatic transmission control - Torque convertor lock up control - Differential and traction control Text / Reference: T/R Book Title/Author T1 U.Kienck, and L. Nielsen, Automotive Control Systems, SAE and Springer-Verlag 2000 LjuboVlacic, Michel Parent, Fumio Harashima, "Intelligent Vehicle Technologies", R1 Butterworth-Heinemann publications, Oxford, 2001 Crouse, W.H & Anglin, D.L,"Automotive mechanics", Intl.Student edition, R2 9thedition, TMH, New Delhi, 2002. William B. Ribbens- Understanding Automotive Electronics, 5th edition, Butter worth R3 Heinemann Woburn 1998. R4 Bosh, "Automotive Handbook", 6th edition, SAE, 2004 Online Resources: Sl.No Website Link
<https://www.electronicsforu.com/technology-trends/tech-focus/automobile-1-industry-sensor> 2 <https://www.youtube.com/watch?v=jyQuRgQHGCK> 3 <https://www.youtube.com/watch?v=8e4QB1DYYJU> 4 <https://www.youtube.com/watch?v=FpGGpgSEU94> 5 <https://www.youtube.com/watch?v=PFNe7NyJeb0>**

5 Understanding control Integrated engine control system - Secondary air Management - Evaporative emission canister purge - Automatic system adjustment - System diagnosis - Improvements in Electronic Engine Control system - Oxygen sensor improvements - Heated exhaust gas oxygen sensor - Fuel injection Timing - Transmission control - Automatic transmission control - Torque convertor lock up control - Differential and traction control Text / Reference: T/R Book Title/Author T1 U.Kienck, and L. Nielsen, Automotive Control Systems, SAE and Springer-Verlag 2000 LjuboVlacic, Michel Parent, Fumio Harashima, "Intelligent Vehicle Technologies", R1 Butterworth-Heinemann publications, Oxford, 2001 Crouse, W.H & Anglin, D.L,"Automotive mechanics", Intl.Student edition, R2 9thedition, TMH, New Delhi, 2002. William B. Ribbens-Understanding

Automotive Electronics, 5th edition, Butter worth R3 Heinemann Woburn 1998. R4 Bosh, "Automotive Handbook", 6th edition, SAE,2004 Online Resources: Sl.No Website Link <https://www.electronicsforu.com/technology-trends/tech-focus/automobile-1-industry-sensor> 2 <https://www.youtube.com/watch?v=jyQuRgQHGck> 3 <https://www.youtube.com/watch?v=8e4QB1DYYJU> 4 <https://www.youtube.com/watch?v=FpGGpgSEU94> 5 <https://www.youtube.com/watch?v=PFNe7NyJeb0> is an official topic in Module 4 of Automotive Power Train. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 5 Understanding control Integrated engine control system
 - Secondary air Management - Evaporative emission canister purge - Automatic system adjustment - System diagnosis - Improvements in Electronic Engine Control system - Oxygen sensor improvements - Heated exhaust gas oxygen sensor - Fuel injection Timing - Transmission control - Automatic transmission control - Torque convertor lock up control - Differential and traction control
- Text / Reference: T/R Book Title/Author T1 U.Kienck, and L. Nielsen, Automotive Control Systems, SAE and Springer-Verlag 2000 LjuboVlacic, Michel Parent, Fumio Harashima, "Intelligent Vehicle Technologies", R1 Butterworth-Heinemann publications, Oxford, 2001 Crouse, W.H & Anglin, D.L,"Automotive mechanics", Intl.Student edition, R2 9thedition,TMH,NewDelhi,2002. William B. Ribbens- Understanding Automotive Electronics, 5th edition, Butter worth R3 Heinemann Woburn 1998. R4 Bosh, "Automotive Handbook", 6th edition, SAE,2004 Online Resources: Sl.No Website Link

[https://www.electronicsforu.com/technology-trends/tech-focus/automobile-1 industry-sensor 2](https://www.electronicsforu.com/technology-trends/tech-focus/automobile-1-industry-sensor-2) <https://www.youtube.com/watch?v=jyQuRgQHGck> 3
<https://www.youtube.com/watch?v=8e4QB1DYYJU> 4
<https://www.youtube.com/watch?v=FpGGpgSEU94> 5
<https://www.youtube.com/watch?v=PFNe7NyJeb0> using standard technical terminology.

- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 5 understanding control integrated engine control system - secondary air management - evaporative emission canister purge - automatic system adjustment - system diagnosis - improvements in electronic engine control system - oxygen sensor improvements - heated exhaust gas oxygen sensor - fuel injection timing - transmission control - automatic transmission control - torque convertor lock up control - differential and traction control text / reference: t/r book title/author t1 u.kienck, and l. nielsen, automotive control systems, sae and springer-verlag 2000 ljubovlacic, michel parent, fumio harashima, "intelligent vehicle technologies", r1 butterworth-heinemann publications, oxford, 2001 crouse, w.h & anglin, d.l, "automotive mechanics", intl.student edition, r2 9th edition, tmh, newdelhi, 2002. william b. ribbens-understanding automotive electronics, 5th edition, butter worth r3 heinemann woburn 1998. r4 bosh, "automotive handbook", 6th edition, sae, 2004 online resources: sl.no website link [https://www.electronicsforu.com/technology-trends/tech-focus/automobile-1 industry-sensor 2](https://www.electronicsforu.com/technology-trends/tech-focus/automobile-1-industry-sensor-2) <https://www.youtube.com/watch?v=jyqurgqhgck> 3 <https://www.youtube.com/watch?v=8e4qb1dyyju> 4 <https://www.youtube.com/watch?v=fpggpgseu94> 5 <https://www.youtube.com/watch?v=pfne7nyjeb0> should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What 5 Understanding control Integrated engine control system - Secondary air Management - Evaporative emission canister purge - Automatic system adjustment - System diagnosis - Improvements in Electronic Engine Control system - Oxygen sensor improvements - Heated exhaust gas oxygen sensor - Fuel injection Timing - Transmission control - Automatic transmission control - Torque convertor lock up control - Differential and traction control Text / Reference: T/R Book Title/Author T1 U.Kienck, and L. Nielsen, Automotive Control Systems, SAE and Springer-Verlag 2000 LjuboVlacic, Michel Parent, Fumio Harashima, "Intelligent Vehicle Technologies", R1 Butterworth-Heinemann publications, Oxford, 2001 Crouse, W.H & Anglin, D.L,"Automotive mechanics", Intl.Student edition, R2 9thedition,TMH,NewDelhi,2002. William B. Ribbens-Understanding Automotive Electronics, 5th edition, Butter worth R3 Heinemann Woburn 1998. R4 Bosh, "Automotive Handbook", 6th edition, SAE,2004 Online Resources: SI.No Website Link https://www.electronicsforu.com/technology-trends/tech-focus/automobile-1-industry-sensor 2 https://www.youtube.com/watch?v=jyQuRgQHGck 3 https://www.youtube.com/watch?v=8e4QB1DYYJU 4 https://www.youtube.com/watch?v=FpGGpgSEU94 5 https://www.youtube.com/watch?v=PFNe7NyJeb0 means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 4-□□ 5 Understanding control Integrated engine control system - Secondary air Management - Evaporative emission

canister purge - Automatic system adjustment - System diagnosis - Improvements in Electronic Engine Control system - Oxygen sensor improvements - Heated exhaust gas oxygen sensor - Fuel injection Timing - Transmission control - Automatic transmission control - Torque convertor lock up control - Differential and traction control Text / Reference: T/R Book Title/Author T1 U.Kienck, and L. Nielsen, Automotive Control Systems, SAE and Springer-Verlag 2000 LjuboVlacic, Michel Parent, Fumio Harashima, "Intelligent Vehicle Technologies", R1 Butterworth-Heinemann publications, Oxford, 2001 Crouse, W.H & Anglin, D.L,"Automotive mechanics", Intl.Student edition, R2 9th edition, TMH, New Delhi, 2002. William B. Ribbens- Understanding Automotive Electronics, 5th edition, Butter worth R3 Heinemann Woburn 1998. R4 Bosh, "Automotive Handbook", 6th edition, SAE, 2004 Online Resources: Sl.No Website Link
1 <https://www.electronicsforu.com/technology-trends/tech-focus/automobile->
2 <https://www.youtube.com/watch?v=jyQuRgQHGCK>
3 <https://www.youtube.com/watch?v=8e4QB1DYYJU>
4 <https://www.youtube.com/watch?v=FpGGpgSEU94>
5 <https://www.youtube.com/watch?v=PFNe7NyJeb0> definition/meaning . application . Technical terms English-

► **Official syllabus detail and terminology (expand in digital version)**

Module study routine: Read the source wording, make a one-page concept map, practise one short answer and one structured answer, then review the Malayalam support notes.

RAPID REFERENCE**Concept and formula bank****Answer structure**

Definition → Principle →
Components/steps →
Application → Conclusion

Use this sequence for descriptive questions when the topic does not require a numerical formula.

Practical record

Aim → Apparatus → Procedure
→ Observation → Result →
Safety

Use this sequence for laboratory, workshop and field activities.

RAPID REVISION**Diagram and source-transcript library****Module 1 source and topic cards**

Jump to official terminology, topic explanations and study guidance.

Module 2 source and topic cards

Jump to official terminology, topic explanations and study guidance.

Module 3 source and topic cards

Jump to official terminology, topic explanations and study guidance.

Module 4 source and topic cards

Jump to official terminology, topic explanations and study guidance.

OFFICIAL SELF-LEARNING**Activities from the syllabus**

Use the extracted official activity wording as the record of required self-learning work.

The official PDF does not expose a separate activities heading in the extracted text; consult the source PDF pages for the exact activity list.

EXAMINATION PREPARATION

Assessment methodology

The following source extract preserves the assessment information detected in the official PDF. Verify mark conversions and institutional updates against the current source before an examination.

The official assessment structure is reproduced from the supplied PDF.

Practice-paper notice: The model paper below is generated for revision and is not an official examination paper.

ACTIVE RECALL

Module-wise questions and answers

These are practice questions based on official topic coverage, not official examination questions.

Module 1

► **Define or explain Outline the functions of engine and controls. (3 marks)**

► **Define or explain Outline the terms related to engine performance. (3 marks)**

► **Define or explain Explain the mapping of engine. 3 Understanding M1. 04 Outline electronic fuel control system 4 Understanding The concept of electronic Engine control system - Engine functions and control - Parameters and variables - Input to controller - output from controller - Definition of engine performance terms - Power - torque - fuel consumption - Volumetric efficiency - Thermal efficiency - calibration, Engine mapping-Effect of air fuel ratio, spark timing and exhaust gas recirculation on performance - Electronic fuel control system - Engine control sequence - open loop control - closed loop control - closed loop operation - Idle speed control. Identify various sensors and actuators used in electronic engine control. (3 marks)**

Module 2

► **Define or explain Outline the engine Variables. (3 marks)**

► **Define or explain Illustrate the working of various sensors. (3 marks)**

► **Define or explain Illustrate the working of various actuators. (3 marks)**

► **Define or explain Identify various actuators and actuators 1 Applying Typical electronic Engine Control System. Variables to be measured, Sensors and actuators- Temperature sensor- Pressure sensor-Mass air flow sensor-Oxygen sensor- Position sensor-Position/speed sensor-Knock sensor- Automotive Engine Control Actuators-Solenoid-switches-relays-motors-Fuel injection- Exhaust gas recirculation actuators- Variable valve timing.. (3 marks)**

Module 3

► **Define or explain 5 Understanding control systems.. (3 marks)**

► **Define or explain Explain various modes for fuel control. (3 marks)**

► **Define or explain Explain Electronic ignition control 5 Understanding Digital power train control systems - features - Components - Digital engine control system diagram - Modes for fuel control - Control during Engine crank - Engine Warm up - After the engine warm up - open loop - closed loop - Acceleration enrichment - Deceleration leaning - Idle speed control - EGR Control - Variable valve timing control - Electronic ignition control-Distributor less Ignition control. Outline the integrated engine control system and automatic transmission. (3 marks)**

Module 4

► **Define or explain Explain integrated engine control systems 4 Understanding Explain improvements in Electronic engine. (3 marks)**

► **Define or explain 5 Understanding control system Explain automatic transmission and traction. (3 marks)**

► **Define or explain 5 Understanding control Integrated engine control system - Secondary air Management - Evaporative emission canister purge - Automatic system adjustment - System diagnosis - Improvements in Electronic Engine Control system - Oxygen sensor improvements - Heated exhaust gas oxygen sensor - Fuel injection Timing - Transmission control - Automatic transmission control - Torque convertor lock up control - Differential and traction control** Text /

Reference: T/R Book Title/Author T1 U.Kienck, and L. Nielsen, Automotive Control Systems, SAE and Springer-Verlag 2000 LjuboVlacic, Michel Parent, Fumio Harashima, "Intelligent Vehicle Technologies", R1 Butterworth-Heinemann publications, Oxford, 2001 Crouse, W.H & Anglin, D.L,"Automotive mechanics", Intl.Student edition, R2 9thedition,TMH,NewDelhi,2002. William B. Ribbens-Understanding Automotive Electronics, 5th edition, Butter worth R3 Heinemann Woburn 1998. R4 Bosh, "Automotive Handbook", 6th edition, SAE,2004

Online Resources: Sl.No Website Link

https://www.electronicsforu.com/technology-trends/tech-focus/automobile- 1 industry-sensor 2 https://www.youtube.com/watch?v=jyQuRgQHGck 3 https://www.youtube.com/watch?v=8e4QB1DYYJU 4 https://www.youtube.com/watch?v=FpGGpgSEU94 5 https://www.youtube.com/watch?v=PFNe7NyJeb0. (3 marks)

PRACTICE ONLY

Practice Model Paper — Not an Official Examination Paper

Attempt one short definition, one structured explanation and one long answer from every module. Use the official assessment pattern reproduced above when selecting time and marks.

Module 1

1-mark definition: State the meaning of **Outline the functions of engine and controls.**

3-mark explanation: Explain one principle, process or comparison from this module.

7-mark answer: Write a structured answer using a labelled diagram, table, procedure, example or application.

Module 2

1-mark definition: State the meaning of **Outline the engine Variables.**

3-mark explanation: Explain one principle, process or comparison from this module.

7-mark answer: Write a structured answer using a labelled diagram, table, procedure, example or application.

Module 3

1-mark definition: State the meaning of **5 Understanding control systems..**

3-mark explanation: Explain one principle, process or comparison from this module.

Module 4

1-mark definition: State the meaning of **Explain integrated engine control systems 4 Understanding Explain improvements in Electronic engine.**

3-mark explanation: Explain one principle, process or comparison from

7-mark answer: Write a structured answer using a labelled diagram, table, procedure, example or application.

this module.

7-mark answer: Write a structured answer using a labelled diagram, table, procedure, example or application.

QUICK REVISION

Exam checklist

Before writing

- Read the command word: define, explain, compare, draw, calculate, analyse or design.
- Use the exact course terminology from the source transcript.
- Plan labels, units, sequence and marks before writing.

Before submitting

- Check every module has been revised.
- Check diagrams, tables, formula symbols and units.
- Separate official syllabus information from personal elaboration.

REFERENCE VOCABULARY

Glossary

Study glossary

Term	Meaning in this lesson
Course Outcome	A capability the student should demonstrate after completing the course.
Module	An official syllabus unit with defined topics and hours.
CIE	Continuous Internal Evaluation.
ESE	End Semester Examination.
Source transcript	A preserved extract of the official syllabus used for coverage checking.

SOURCE-SUPPORTED REFERENCES

References and web resources

References listed in the official PDF

References are included in the official source PDF.

Web resources listed in the official PDF

- Sl.No Website Link
- <https://www.electronicsforu.com/technology-trends/tech-focus/automobile-industry-sensor> <https://www.youtube.com/watch?v=jyQuRgQHGCK>
<https://www.youtube.com/watch?v=8e4QB1DYYJU>
<https://www.youtube.com/watch?v=FpGGpgSEU94>
<https://www.youtube.com/watch?v=PFNe7NyJeb0>

IN-PAGE SEARCH**Search this lesson**

Prepared from the supplied official SITTTTR syllabus PDF for Course 5053A. Verify institutional updates against the current official curriculum.